جامعة سوهاج
Sohag University



LEX

2025-2026

CS DEPARTMENT
GRADUATION PROJECT

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Contents

Abstract 6

Chapter 1: Introduction 7

Chapter 2: Problem Definition 10

Chapter 3: Project Analysis 23

Chapter 4: Project Tools 47

Chapter 5: System Design 59

Chapter 6: System Testing 70

Chapter 7: Conclusion and discussion 75

Chapter 8: Bibliography 77

List of Figures

Figure 1. Agile SDLC Phases..... 25

Figure 2. Agile SDLC iterative cycles (sprints)..... 26

Figure 3. User System Use Case Diagram..... 37

Figure 4. Lawyer System Use Case Diagram..... 44

Figure 5. Admin System Use Case Diagram..... 49

Figure 6. Entity Relationship Diagram (ERD)..... 52

Figure 7. Data Flow Diagram Level Zero 53

Figure 7. Data Flow Diagram Level One..... 54

List of Tables

Table 1: User Issues..... 12

Table 2: Lawyer and Law Firm Issues Page..... 14

Table 3: Existing Systems Issues Page..... 16

Table 4: User-Focused Objectives..... 17

Table 5: Lawyer and Law Firm-Focused Objectives..... 19

Table 6: System-Focused Objectives..... 20

List of Figures

Table 7: Software Requirements..... 21

Abbreviations

Symbol	Expression
SDLC	Software Development Lifecycle
ERD	Entity-Relationship Diagram
DFD	Data Flow Diagram
FK	Foreign Key
DS	Data Store
I/O	Input / Output
HTTP	Hypertext Transfer Protocol
SMTP	Simple Mail Transfer Protocol
SQL	Structured Query Language
RDBMS	Relational Database Management System
VS Code	Visual Studio Code
SDK	Software Development Kit
JIT	Just-In-Time
AOT	Ahead-Of-Time
API	Application Programming Interface
UI	User Interface
UX	User Experience

Abstract

LEX is an innovative digital platform designed to modernize and streamline the legal ecosystem for lawyers, law firms, and clients across Egypt. The platform addresses key challenges such as inefficient communication, lack of case transparency, and the difficulty of verifying legal professionals and institutions. By integrating digital identity verification, secure document management, and real-time case tracking, **LEX** enables a more transparent, accessible, and trusted legal environment.

The system connects users, lawyers, and law firms within a unified ecosystem where users can submit legal requests, verify lawyer credentials through ID and license checks, and track case progress via built-in chat and notification systems. Lawyers and law firms can manage client profiles, handle case documentation, and maintain verified professional records including license certificates and bar association IDs. The platform also includes role-based access for administrators and verification officers ("Activators") who oversee and approve lawyer and firm registrations to ensure authenticity.

The **LEX Project** follows an agile and iterative development approach, emphasizing adaptability and user-centric design. The multidisciplinary team—comprising backend, frontend, Flutter, and UI/UX developers—leveraged modern technologies such as **.NET** for backend services, **React** for the web interface, **Flutter** for cross-platform mobile application development, **PostgreSQL** for data management, and **Figma** for UI/UX design. Comprehensive system documentation, including **Entity-Relationship Diagrams (ERD)**, **Use Case Diagrams**, and **Activity Diagrams**, ensures alignment between technical and non-technical stakeholders throughout the development process.

Abstract

Ultimately, **LEX** aims to digitally transform the legal sector in Egypt by fostering transparency, improving accessibility to legal services, and empowering professionals through technology. By bridging the gap between clients and verified legal practitioners, the platform contributes to building a more reliable, efficient, and equitable justice ecosystem.

Chapter 1
Introduction

Introduction

Egypt's legal ecosystem, particularly in rural and semi-urban regions, faces major challenges rooted in outdated administrative processes, limited digital infrastructure, and weak verification mechanisms. Many citizens struggle to access reliable legal services due to the lack of transparency, inefficient communication between clients and lawyers, and the prevalence of unverified practitioners. Lawyers and law firms, in turn, face difficulties in managing client information, authenticating credentials, and maintaining secure communication channels. While digital transformation has reshaped sectors such as finance and education, the legal field remains largely paper-based and fragmented, creating barriers to accessibility, trust, and efficiency.

LEX emerges as a comprehensive digital platform designed to bridge these gaps by connecting users, lawyers, and law firms within a secure, transparent, and user-friendly ecosystem. The platform digitizes essential legal workflows—such as lawyer verification, case management, and client communication—while ensuring data protection and regulatory compliance. Through role-based access, the system empowers administrators and verification officers ("Activators") to validate legal professionals and firms before approval. Additionally, built-in features like encrypted chat, digital document sharing, and real-time status updates simplify case tracking and communication. By combining accessibility with accountability, LEX seeks to redefine how individuals and professionals interact within Egypt's legal system.

Lawyers and law firms benefit from integrated digital tools that streamline daily operations and enhance client service quality. Intelligent case management and scheduling systems help optimize workload distribution and reduce administrative effort, allowing legal professionals to focus more on advocacy and legal strategy. Transparent performance tracking, digital documentation, and secure communication channels empower lawyers to build stronger client relationships and maintain data integrity. Automated record-keeping, document generation, and compliance tools further minimize paperwork, reduce errors, and ensure adherence to legal and ethical standards.

Beyond efficiency, **LEX** aims to transform Egypt's legal landscape by digitizing and formalizing manual workflows. The platform enables lawyers, firms, and clients to operate within a transparent, verified, and accountable ecosystem. Through verified profiles, digital case records, and structured feedback, LEX strengthens trust, professionalism, and client satisfaction across the legal sector.

LEX is more than a platform—it's a commitment to redefining legal services through justice, transparency, and innovation. With user-focused design, verified partnerships, and scalable technology, LEX transforms Egypt's fragmented legal processes into a connected and efficient digital justice system.

Chapter 2
Problem Definition

Problem Definition

Egypt's legal sector continues to face major challenges due to the absence of unified digital systems and standardized verification processes. Most legal interactions between clients, lawyers, and law firms remain paper-based, time-consuming, and prone to human error. This leads to delayed communication, misplaced documents, and limited transparency in case handling.

These issues reduce public trust in legal services and limit accessibility—particularly in rural and semi-urban areas. There is a growing need for a comprehensive digital platform that ensures verification, secure communication, and efficient management of legal processes. **LEX** addresses this need by providing a transparent, user-centric system that connects verified lawyers, law firms, and clients in a secure digital environment.

This section highlights the **issues and challenges** faced by **ordinary users and legal professionals**, as well as the **limitations of existing systems** that our project aims to address and resolve.

● User Issues

Table 1: User Issues

Issues	Weight
1. Difficulty Finding Verified Lawyers or Law Firms	10
2. Lack of Trust and Verification System	9
3. Poor Communication with Legal Representatives	8
4. No Real-Time Case or Issue Updates	8
5. Limited Access to Legal Help in Remote Areas	7

---

1. **Lack of Reliable and Organized Legal Services**:
    - Citizens struggle to find verified, reliable legal representation.
    - No consistent directory or trusted listing of lawyers and law firms.

2. **Difficulty Accessing Essential Legal Assistance**:
    - People in rural and semi-urban areas face barriers

to reach lawyers or law firms at the right time.

- Lack of professional remote consultation and document delivery increases delays and costs.

3. **No Secure and Standardized Verification System**
   - Hard to validate lawyer credentials, bar membership, or firm licenses reliably.
   - High risk of misinformation, fraud, or unqualified representation.
4. **Poor Client–Lawyer Communication and Collaboration**
   - No secure, auditable chat and file-exchange mechanism tied to cases.
   - Fragmented communication across phone calls, WhatsApp, and paper letters.
5. **Limited Access to Legal Help in Remote Areas**
   - People in rural or low-connectivity regions cannot reach professional legal assistance easily.
   - Absence of online consultations or lightweight access tools.

● Lawyer and Law Firm Issues

Table 2: Lawyer and Law Firm Issues

Issues	Weight
1. Low Visibility and Client Reach	10
2. Verification and Reputation Challenges	9
3. Inefficient Client Communication	8
4. Absence of Centralized Case Notifications	7
5. Limited Digital Presence in Remote Regions	6

1. **Low Visibility and Client Reach**:
   - Lawyers and law firms struggle to attract new clients due to poor online presence.
   - No centralized system connects professionals with people seeking verified legal support.
2. **Verification and Reputation Challenges**:
   - Professionals cannot showcase verified credentials or licenses digitally.
   - Unverified listings reduce user trust and engagement.
3. **Inefficient Client Communication**:
   - Communication with clients is unstructured, often across multiple platforms.
   - No single, secure medium for legal correspondence or document sharing.

4. **Absence of Centralized Case Notifications**:
    - Lawyers must manually inform clients about case progress or deadlines.
    - Lack of automatic reminders leads to missed updates and reduced efficiency.
5. **Limited Digital Presence in Remote Regions**:
    - Small firms lack tools to provide digital consultations or maintain visibility in rural areas.
    - Absence of online booking and communication tools limits professional growth.

● Existing Systems Issues

Table 3: Existing Systems Issues

Issues	Weight
1. Lack of Unified Lawyer-User Platform	9
2. Weak Verification and Security Mechanisms	8
3. Limited Notification and Tracking Systems	7
4. Poor User Experience and Accessibility	7

1. **Lack of Unified Lawyer-User Platform**:
    - Existing services are fragmented, with no platform providing end-to-end connection between users, lawyers, and law firms.
2. **Weak Verification and Security Mechanisms**:
    - Current systems lack proper validation of lawyer identities, licenses, and firm registrations.
    - This creates a high risk of misinformation or fraudulent profiles.
3. **Limited Notification and Tracking Systems**:
    - Users receive no automated updates regarding their case or communication status.
    - Manual follow-up causes delays and dissatisfaction.
4. **Poor User Experience and Accessibility**:
    - Outdated designs and lack of cross-platform support make interaction difficult.
    - Low responsiveness and limited mobile optimization reduce adoption.

Objectives

This section outlines the **core objectives of the LEX platform**, which aims to overcome the challenges faced by users and legal professionals, as well as the weaknesses of current systems. Each objective provides a focused solution that enhances reliability, efficiency, and accessibility in legal service connections.

● User-Focused Objectives

Table 4: User-Focused Objectives

1. Simplify Lawyer and Law Firm Discovery
2. Build a Verified and Transparent Legal Network
3. Enable Real-Time Case and Issue Updates
4. Improve Communication with Lawyers
5. Enhance Accessibility for All Users

1. **Simplify Lawyer and Law Firm Discovery**:
    - Develop an intelligent search system where users can easily find verified lawyers or firms based on specialty, location, and availability.
2. **Build a Verified and Transparent Legal Network**:
    - Implement ID and license verification during registration to ensure authenticity.
    - Display verification badges to improve trust and transparency.

3. **Enable Real-Time Case and Issue Updates**:
    - Integrate automated notifications that inform users about case status, lawyer responses, or document updates instantly.
4. **Improve Communication with Lawyers**:
    - Create a secure, case-linked chat feature for exchanging messages and files between users and lawyers.
5. **Enhance Accessibility for All Users**:
    - Provide a mobile-friendly and low-data interface to ensure usability in all regions, including rural areas.

● Lawyer and Law Firm-Focused Objectives

Table 5: Lawyer and Law Firm-Focused Objectives

	Objectives
1.	Increase Visibility and Client Connections
2.	Simplify Verification and Reputation Management
3.	Streamline Client Communication
4.	Automate Notifications and Case Updates
5.	Support Digital Inclusion and Growth

1. **Increase Visibility and Client Connections**:
    - Offer search listing, specialization filters, and rating systems to help verified lawyers reach more clients.
2. **Simplify Verification and Reputation Management**:
    - Allow professionals to upload required documents and display verified status within the platform.
3. **Streamline Client Communication**:
    - Integrate secure in-app chat for managing client interactions efficiently and maintaining communication records.
4. **Automate Notifications and Case Updates**:
    - Enable automatic updates to clients when case status or documents change.
5. **Support Digital Inclusion and Growth**:
    - Design features for smaller law firms to manage profiles, clients, and communications efficient

● System-Focused Objectives

Table 6: System-Focused Objectives

	Objectives
1.	Ensure Secure and Verified Access
2.	Enable Automated Notifications and Tracking
3.	Provide Scalable Architecture for Growth
4.	Deliver User-Centered Experience

1. **Ensure Secure and Verified Access**:
    - Implement encrypted authentication and verification systems to protect user and lawyer data.
2. **Enable Automated Notifications and Tracking**:
    - Build backend logic to deliver instant updates regarding new cases, lawyer approvals, or status changes.
3. **Provide Scalable Architecture for Growth**:
    - Design a scalable platform capable of handling large numbers of users, lawyers, and firms without performance loss.
4. **Deliver User-Centered Experience**:
    - Develop a clean, intuitive interface ensuring seamless navigation and accessibility across devices.

Requirements

1. Software Requirements:

Software	Minimum Requirements
Operating System	IOS, Android
Android	9 and Newer.
IOS	13 and Newer.
Browser	Any browser like Google Chrome, Microsoft Edge, and others.
Internet Connection	Stable Internet Connection

Table 7. Software Requirements

2. Security Requirements:

2.1. Authentication and Authorization: Only authorized users can access certain application areas or perform specific actions. This involves implementing strong authentication mechanisms, such as multi-factor authentication, and clearly defined roles and permissions.

2.2. Secure Session Management: Secure session management prevents issues like session hijacking or fixation. This includes practices such as setting appropriate session timeouts and securely handling session tokens.

3. Efficiency Requirements:

The application will be designed and programmed to ensure users enjoy a seamless and smooth experience. By optimizing performance and minimizing delays, the application will provide a responsive and reliable interaction, enhancing user satisfaction and overall usability.

Constraints

To ensure the effectiveness of the application, and that the application aligns with the target audiences' needs and requirements, several constraints need to be considered while developing the application.

1. **Security & Privacy:**
   The application must adhere to security and privacy standards to safeguard user information, payment details, and other sensitive data. Appropriate measures must be in place to ensure data confidentiality, integrity, and availability.
2. **User Experience:**
   The application should deliver a seamless and user-friendly experience for passengers and drivers. Attention must be given to design, navigation, responsiveness, and accessibility across devices and user roles while maintaining a clear and easy-to-understand system.
3. **Development Time:**
   The initial application must be ready for use and well-tested by June 1st to be submitted as the final project for the senior year.

Recognizing and tackling these challenges in creating a delivery application can help build a system that fulfills user requirements, adheres to regulations, delivers a satisfying, easy-to-use, user experience, and stays within the allocated resources and budget. By proactively addressing these constraints, developers can create a more efficient, reliable, and user-centric solution that meets expectations and enhances overall user and stakeholder satisfaction.

Chapter 3
Project Analysis

Project Analysis

Every software project is unique, demanding a tailored Software Development Life Cycle (SDLC) strategy that aligns with its specific needs, team dynamics, and external conditions. Selecting the right methodology helps accelerate development timelines, strengthen teamwork, and enhance adaptability to evolving requirements.

For our project, we've chosen to adopt an Agile SDLC approach. The Agile workflow and development process is an iterative and incremental approach to software development that emphasizes flexibility, collaboration, and customer satisfaction. It focuses on delivering small, functional increments of a product, known as iterations or sprints, rather than delivering the entire product at once. An Agile workflow and development process consists of the following:

Core Principles of Agile:

Agile is guided by the Agile Manifesto, which emphasizes:

- Individuals and interactions over processes and tools.
- Working software over comprehensive documentation.
- Customer collaboration over contract negotiation.
- Responding to change over following a plan.

Agile Software Development Life Cycle Phases

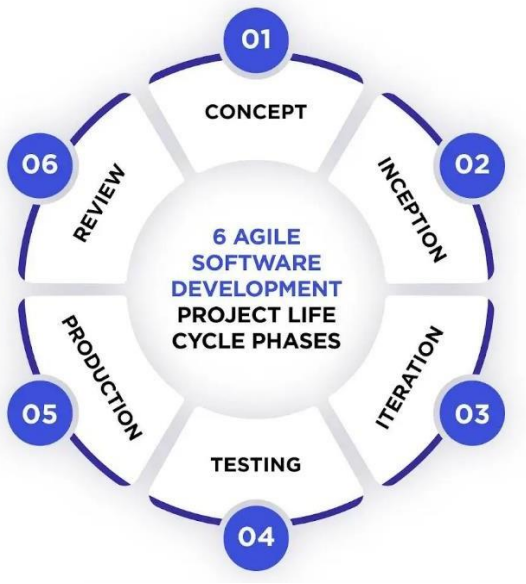


Figure 1. Agile SDLC Phases

1. Concept/Envisioning:

Objective: Define the project vision, scope, and business case.

Activities:

- Identify stakeholders and high-level requirements.
- Assess feasibility, risks, and ROI (Return on Investment).
- Draft a product vision statement and roadmap.

Outcome: A clear understanding of the project's purpose and goals.

2. Inception

Objective: Establish the team, prioritize features, and plan the first iteration.

Activities:

- Form a cross-functional team (developers, testers, designers, Product Owner, Scrum Master).
- Create user stories (customer-centric feature descriptions) and populate the product backlog (prioritized list of requirements).

- Define acceptance criteria and non-functional requirements.

Plan the release timeline and initial sprints.

Outcome: A refined product backlog and a team ready to start development.

3. Iteration/Construction

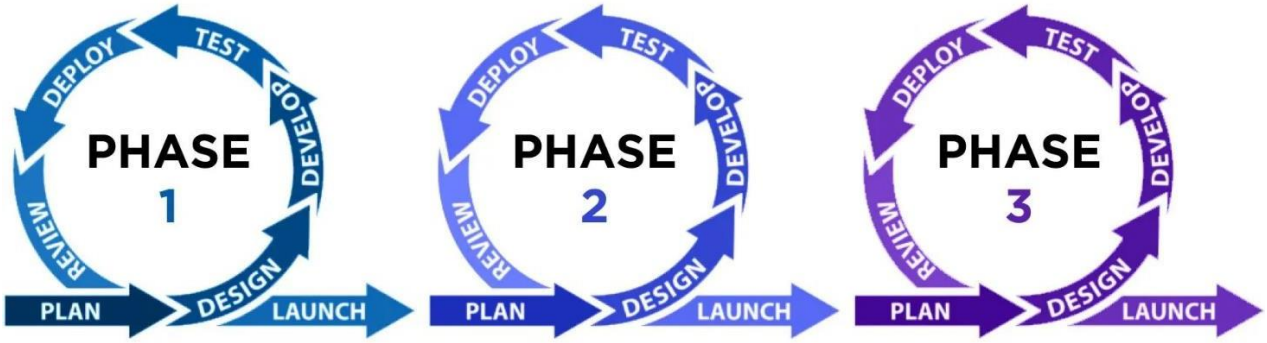


Figure 2. Agile iterative cycles (sprints)

Objective: Deliver working software increments through short, time-boxed cycles (sprints).

Activities:

- Sprint Planning:
  - Select user stories from the product backlog for the sprint (forming the sprint backlog).
  - Break stories into tasks and estimate effort (e.g., using story points or hours).
- Daily Stand-Up:
  - 15-minute meetings to sync on progress, blockers, and daily goals.
- Development & Testing:
  - Code, test, and integrate features daily (using Continuous Integration).
  - Conduct automated, manual, and exploratory testing.
- Sprint Review:

- Demo the increment to stakeholders for feedback.

- Sprint Retrospective:
  - Reflect on team performance and identify improvements.

Outcome: A potentially shippable product increment at the end of each sprint.

4. Release

Objective: Deploy the product to users and ensure it meets quality standards.

Activities:

- Finalize user acceptance testing (UAT) and bug fixes.
- Prepare documentation, training, and release notes.
- Deploy to production (often via Continuous Delivery pipelines).
- Monitor post-release performance and gather user feedback.

Outcome: A functional product in the hands of users.

5. Production & Maintenance

Objective: Support the live product and iterate based on feedback.

Activities:

- Provide technical support and address defects.
- Collect user feedback to refine the product backlog.
- Plan future sprints for enhancements or new features.

Outcome: Continuous improvement and alignment with user needs.

6. Retirement

Objective: Phase out the product when obsolete or replaced.

Activities:

  - Notify users and plan migration to a new system (if applicable).
- Archive code and documentation.

Outcome: Graceful sunset of the product.

Core Agile Practices in the Lifecycle

Iterative Development: Build in small increments (sprints) for rapid feedback.

Continuous Integration/Delivery (CI/CD): Automate testing and deployment.

User Stories & Backlog Refinement: Keep requirements dynamic and prioritized.

Transparency: Use tools like Kanban boards or burndown charts to visualize progress.

Collaboration: Daily stand-ups, pair programming, & stakeholder demos.

Why use Agile SDLC?

Flexibility: Adapt to changing requirements mid-project.

Customer-Centric: Regular feedback ensures the product solves real problems.

Reduced Risk: Early delivery of MVP (Minimum Viable Product) validates assumptions.

Team Empowerment: Self-organizing teams drive innovation and accountability.

Use Case Scenario

Use case name:	Lawyer Registration
Unique ID:	UC-LW-01
Area:	Lawyer System
Actors:	Lawyer
Stakeholders:	Lawyers, System Administrato
Description:	The system allows lawyers to register by entering professional details and verifying their identity.
Triggering Event:	Lawyer wants to create an account
Triggering Type:	External

Steps Performed (Main Path)	**Information for Steps**
1. Open Application	Lawyer opens the application interface.
2. Enter Personal Information	Lawyer enters personal and professional details such as name, email, license number, and specialization.
3. Upload Verification Documents:	Lawyer uploads required legal documents for verification.

Use Case Scenario

4. System Validates Data	The system validates the submitted information and documents.
5. Account Creation	The system approves the lawyer account.
Preconditions: • he lawyer is not registered in the system. • The lawyer has valid legal credentials.	
Postcondition: • The lawyer account is successfully created and activated.	
Assumptions: • The provided documents are authentic. • The system can verify lawyer credentials.	

Use Case Scenario

Use case name:	User Sign Up
Unique ID:	UC-US-01
Area:	User System
Actors:	User
Stakeholders:	Users, System Administrator
Description:	The system allows new users to create an account by entering personal information and verifying their email
Triggering Event:	User wants to create a new account
Triggering Type:	External

Steps Performed (Main Path)	**Information for Steps**
1. Open Application	User opens the application interfac
2. Enter Personal Information	User enters required personal data such as name, email, and password.
3.System Sends Verification Code:	The system sends a verification code to the user's email.
4. Field Owner Logs In	User enters the verification code to confirm the email address

Use Case Scenario

5. Account Creation	The system creates the user account successfully.
Preconditions: • The user is not registered in the system. • The user is a human (not a robot).	
Postcondition: • The user account is successfully created and activated.	
Assumptions: • The system is available and connected to the internet. • Email service is functioning correctly.	

Use Case Scenario

Use case name:	Admin Login
Unique ID:	UC-AD-01
Area:	Admin System
Actors:	Admin
Stakeholders:	System Administrator, Users, Lawyers
Description:	The system allows administrators to log in and access the admin dashboard.
Triggering Event:	Admin wants to access the system
Triggering Type:	External

Steps Performed (Main Path)	**Information for Steps**
1. Open Admin Panel	Admin opens the admin login page.
2. Enter Credentials	Admin enters username and password.
3. System Validates Credentials	The system validates admin login information.
4.Admin Role Verification	The system verifies that the user has admin privileges.

Use Case Scenario

5. Dashboard Access	Admin gains access to the admin dashboard.
Preconditions: • Admin account exists. • Admin has valid credentials.	
Postcondition: • Admin successfully accesses the dashboard.	
Assumptions: • System security services are operational	

Analysis Diagrams

To ensure clarity and alignment across all project phases, we have developed a suite of detailed visual diagrams that comprehensively outline system requirements, workflows, and architectural design. These diagrams—including Entity Relationship Diagram, use case Diagrams, and system and activity diagrams—serve as strategic tools to bridge communication gaps between technical and non-technical stakeholders. By providing a structured visual framework, they enable precise interpretation of functional specifications and reduce ambiguities.

List of Diagrams included:

1. Use Case Diagrams & Scenarios
2. Entity Relationship Diagram (ERD)
3. Data Flow Diagram (DFD)

1. Use Case Diagram & Scenarios

➢ User System

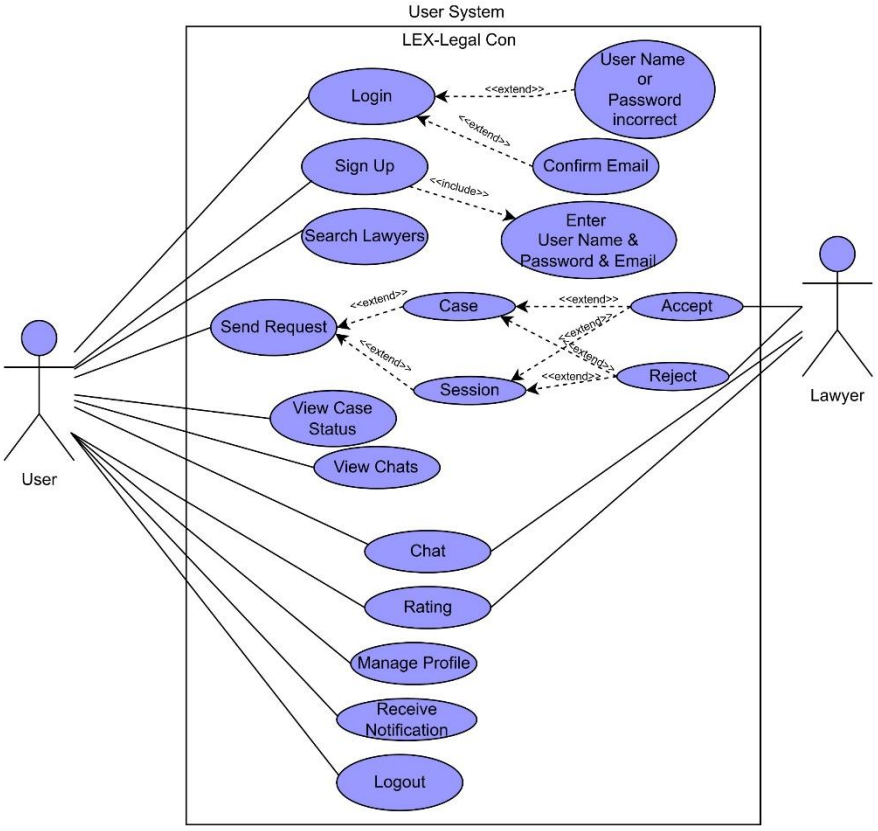


Figure 3. User System Use Case Diagram

Use Case 1: Sign up

- Primary Actor: User, System.
- Precondition: must be human not robot.
- User is not registered .

Main Success Scenario:

- User opens the application.
- User enters personal information.
- System sends verification code.
- User confirms email.
- Account is created successfully.

Exception Scenario:

- Password is weak.
- Email already exists.
- Some sections can't be empty.

Use Case 2: Login

- Primary Actor: User.
- Precondition: Must has account on system.

Main Success Scenario:

- User enters email and password.
- System validates credentials.
- User accesses the system..

Exception Scenario:

- Password is wrong.
- Username or password is incorrect.
- Account is inactive
- Sections can't be empty.

Use Case 3: Search Lawyers

- Primary Actor: User.
- Precondition: The user must be registered in the system and logged in successfully.

Main Success Scenario:

- User enters search criteria.
- System displays lawyers list.
- User filters by specialty.

Exception Scenario:

No lawyers found

Use Case 4: Send Case Request

- Primary Actor: User .
- Precondition: User is logged in.

Main Success Scenario:

- User selects a lawyer.
- User submits case details.
- System sends request to lawyer..

Exception Scenario:

- Missing required information.

Use Case 5: Book Appointment

- Primary Actor: User.
- Precondition: Lawyer is available.

Main Success Scenario:

- User selects date and time.
- System confirms appointment.
- Appointment is saved..

Exception Scenario:

- Selected time unavailable.

Use Case 6: View Case Status

- Primary Actor: User.
- Precondition: Case exists.

Main Success Scenario:

- User opens case details.
- System displays case status..

Exception Scenario:

- Case not found.

Use Case 7: View Chats

- Primary Actor: User.
- Precondition: User is logged in

Main Success Scenario:

- User opens chat list.

System displays conversations.

Use Case 8: Chat

- Primary Actor: User, Lawyer.
- Precondition: Case exists.

Main Success Scenario:

- User opens a chat.
- Messages are exchanged.

Exception Scenario:

- Network error.

Use Case 9 : Rating

- Primary Actor: User.
- Preconditions: Case is completed.

Main Success Scenario:

- .User submits rating.

System saves rating.

Exception Scenario:

- Rating submission failed.

Use Case 10 : Manage Profile

- Primary Actor: User.
- Preconditions:
  - User is logged in.

Main Success Scenario:

- User edits profile.
- System saves changes.

Exception Scenario:

- Invalid data entered.

Use Case 11: Receive Notifications

- Primary Actor: User.
- Preconditions:
  - User account exists.

Main Success Scenario:

- System sends notification.
- User views notification.

Exception Scenario:

- Notification delivery failed.

Use Case 12: Logout

- Primary Actor: User .
- Preconditions:
  - User is logged in.

Main Success Scenario:

- User logs out.
- Session is terminated .

Exception Scenario:

- Logout failed.

➢ Lawyer System

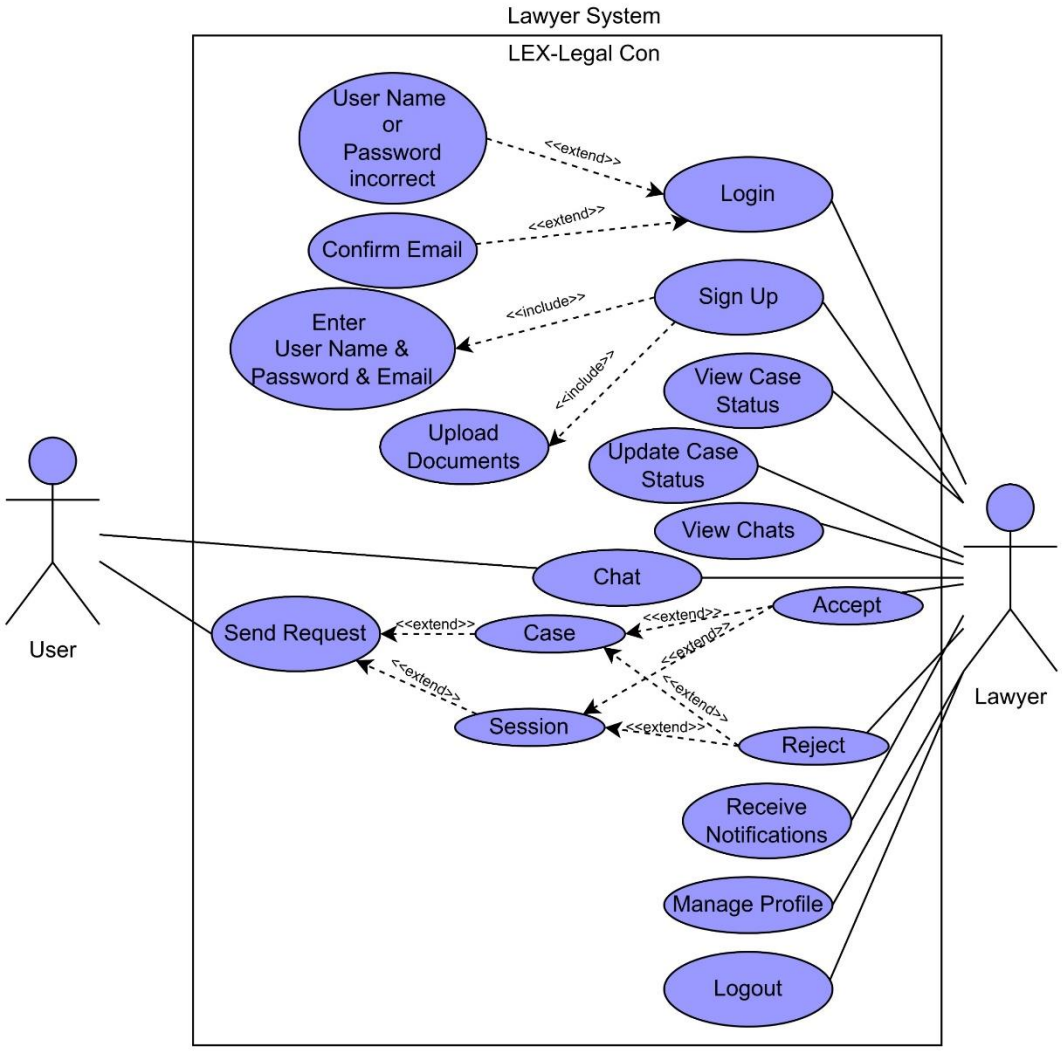


Figure 4. Lawyer System Use Case Diagram

Use Case 1: Sign up

- Primary Actor: Lawyer.
- Precondition: must be human not robot.
- Lawyer is not registered.

Main Success Scenario:

- Lawyer enters personal data.
- Lawyer uploads documents.

➢ **Lawyer System**

- System sends application for review

Exception Scenario:

- Documents missing or invalid.

Use Case 2: Login

- Primary Actor: Lawyer.
- Precondition: must be human not robot.
- Lawyer account approved.

Main Success Scenario:

- Lawyer enters credentials.
- System validates login.

Exception Scenario:

- Incorrect credentials.

Use Case 3: View Case Requests

- Primary Actor: Lawyer.
- Requests exist.

Main Success Scenario:

- Lawyer views incoming requests.

Exception Scenario:

- No requests available.

Use Case 4: Accept Case

- Primary Actor: Lawyer.
- Precondition: Case request exists.

Main Success Scenario:

- Lawyer accepts case.
- System updates case status.

Exception Scenario:

- Request expired.

Use Case 5: Reject Case

- Primary Actor: Lawyer .
- Precondition: Case request exists.

Main Success Scenario:

- Lawyer rejects case.
- System updates case status.

Exception Scenario:

- Request already handled.

Use Case 6: Update Case Status

- Primary Actor: Lawyer.
- Preconditions:
  - Case accepted.

Main Success Scenario:

- Lawyer updates progress.
- System saves status.

Exception Scenario:

- Update failed .

Use Case 7: Manage Appointments

- Primary Actor: Lawyer.
- Precondition: Appointments exist.

Main Success Scenario:

- Lawyer views appointments.
- Lawyer confirms or reschedules.

Exception Scenario:

- Appointment conflict.

Use Case 8: View Chats

- Primary Actor: Lawyer.
- Preconditions: Lawyer logged in.

Main Success Scenario:

- Lawyer opens chat list.

Exception Scenario:

- No chats available.

Use Case 9: Receive Notifications

- Primary Actor: Lawyer.
- Preconditions: Lawyer account active.

Main Success Scenario:

- System sends notification.
- Lawyer views notification.

Exception Scenario:

- Notification failed.

Use Case 10: Manage Profile

- Primary Actor: Lawyer.
- Preconditions: Lawyer logged in.

Main Success Scenario:

- Lawyer updates profile.

Exception Scenario:

- Invalid data.

Use Case 11: Logout

- Primary Actor: User.
- Preconditions:User is logged in.

Main Success Scenario:

- User logs out.
- Session is terminated.

Exception Scenario:

Logout failed.

➢ Admin System

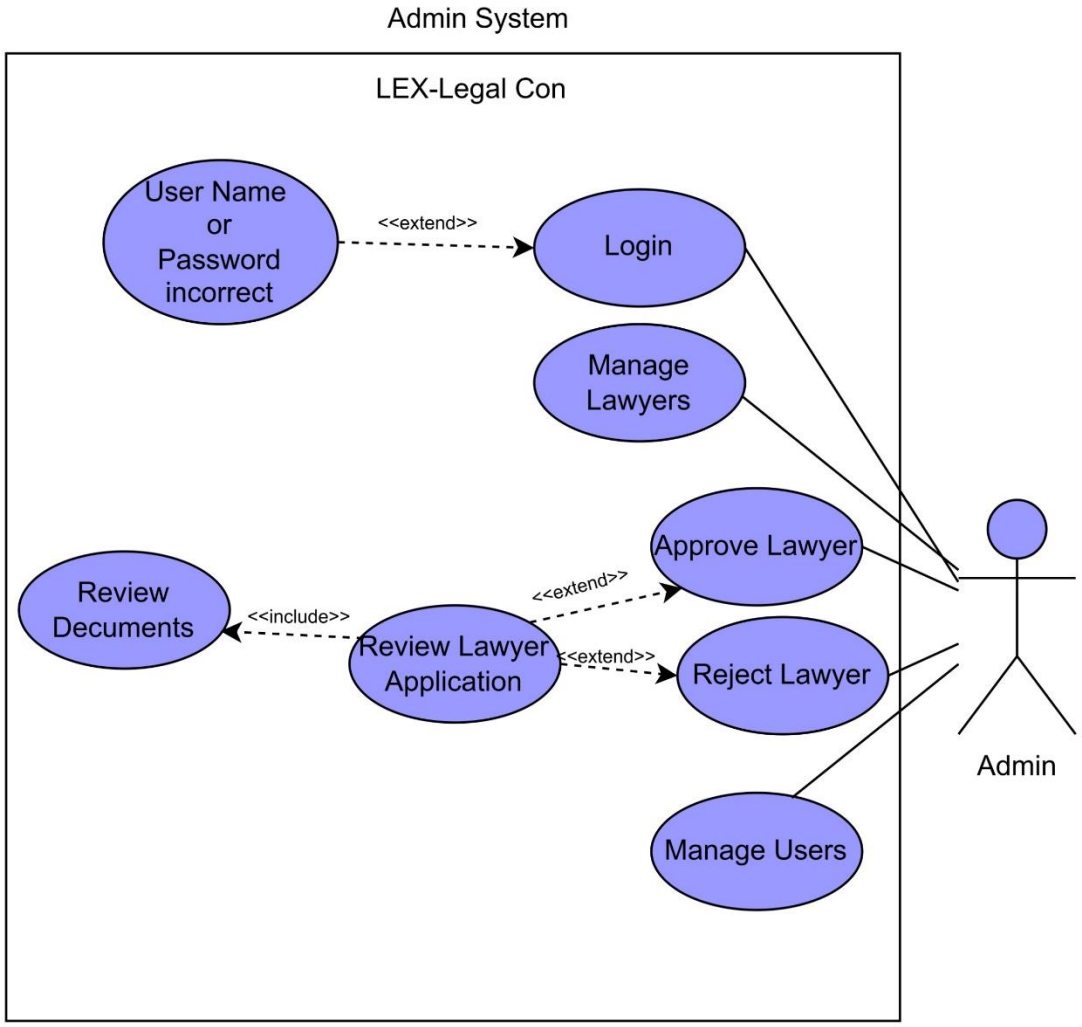


Figure 5. Admin System Use Case Diagram

Use Case 1: Review Lawyer Application

- Primary Actor: Admin.
- Precondition: Application exists.

Main Success Scenario:

- Admin reviews application.
- Admin checks documents.

Exception Scenario:

- Incomplete application.

Use Case 2: Review Documents

- Primary Actor: Admin.
- Precondition: Documents submitted.

Main Success Scenario:

- Admin reviews documents.

Exception Scenario:

- Documents invalid.

Use Case 3: Approve Lawyer

- Primary Actor: Admin.
- Precondition: Application reviewed.

Main Success Scenario:

- Admin approves lawyer.
- Account activated.

Exception Scenario:

- Approval failed.

Use Case 4: Reject Lawyer

- Primary Actor: Admin.
- Precondition: Application reviewed.

Main Success Scenario:

- Admin rejects lawyer.

Exception Scenario:

- Rejection failed.

Use Case 5: Manage Lawyers

- Primary Actor: Admin.
- Precondition: Lawyers exist.

Main Success Scenario:

- Admin manages lawyer accounts.

Exception Scenario:

- Operation failed.

Use Case 5: Manage Users

- Primary Actor: Admin.
- Precondition: Users exist.

Main Success Scenario:

- Admin manages Users accounts.

Exception Scenario:

- Operation failed.

Entity Relationship Diagram (ERD)

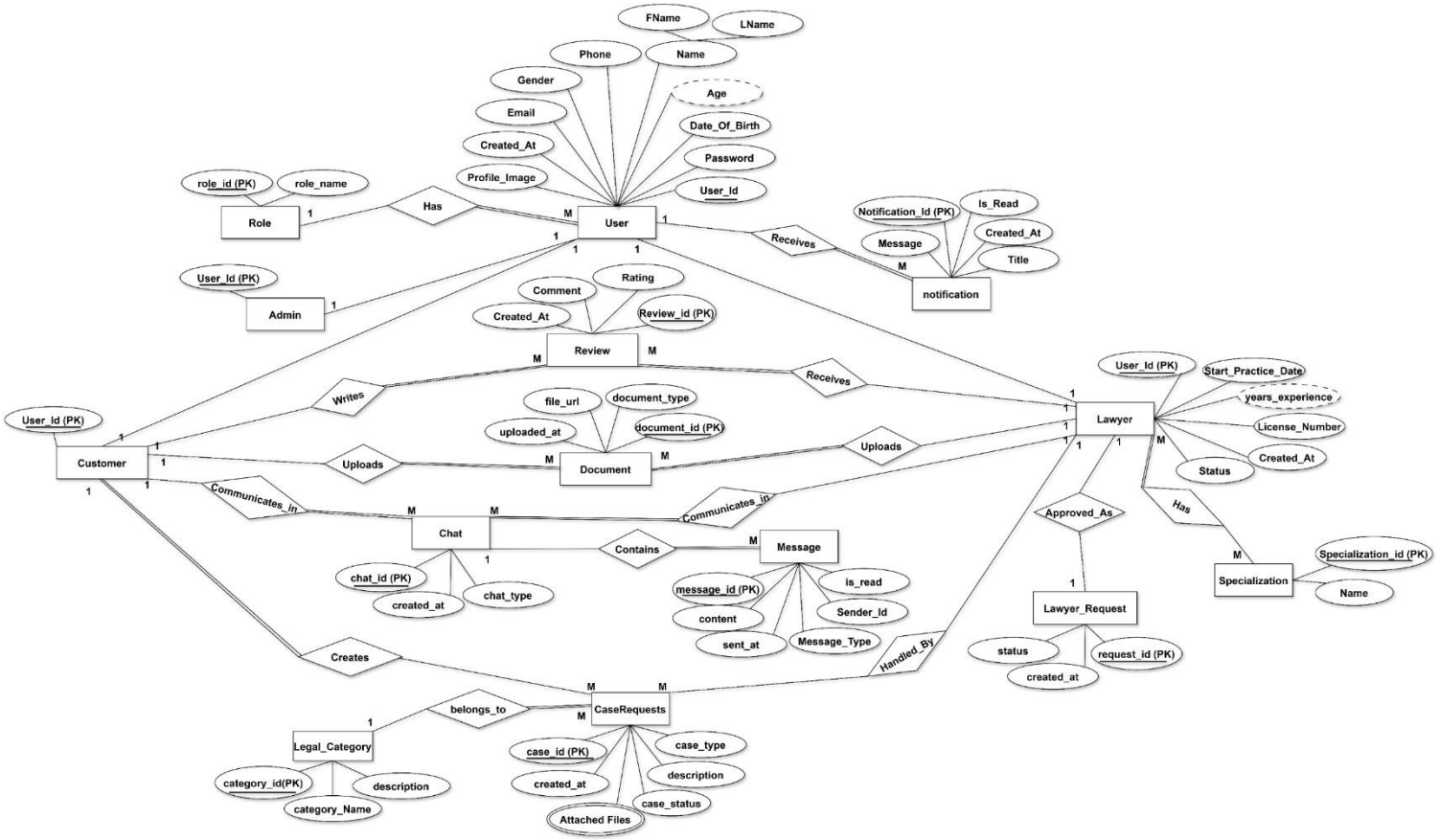


Figure 6. Entity Relationship Diagram (ERD)

3. Data Flow Diagram (DFD)

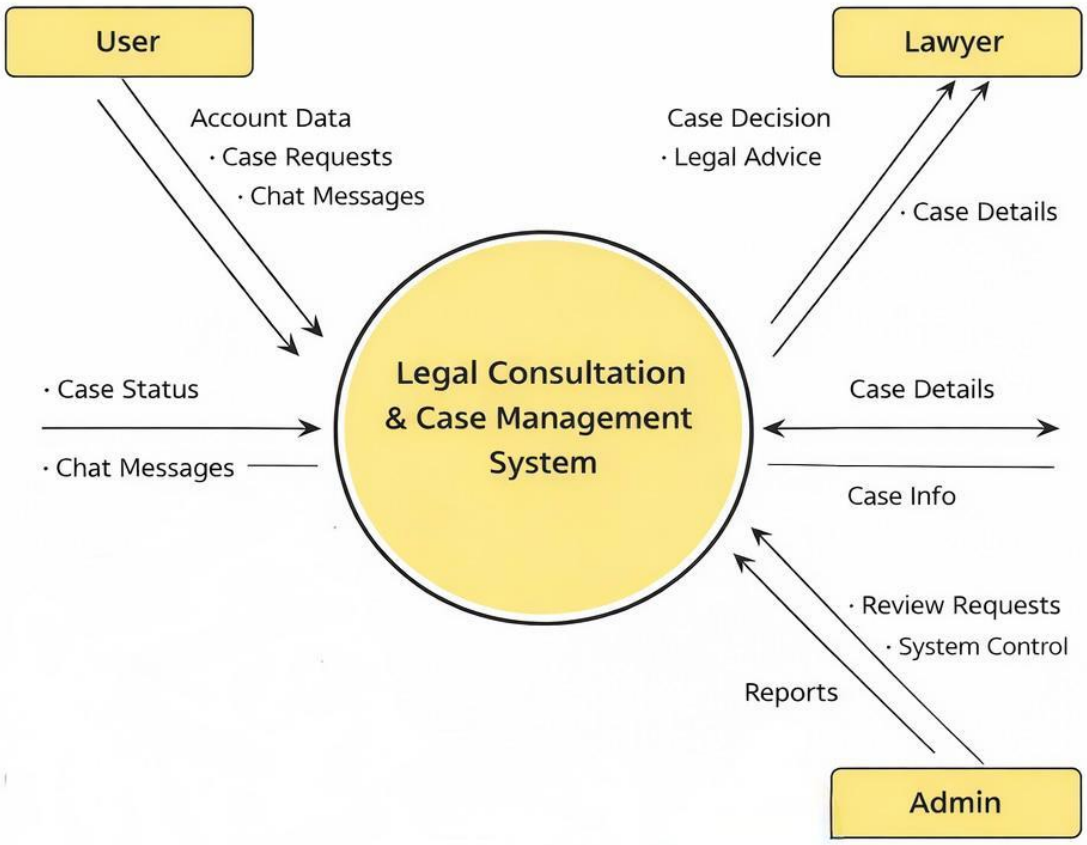


Figure 7. Data Flow Diagram (DFD) LEVEL 0

3. Data Flow Diagram (DFD)

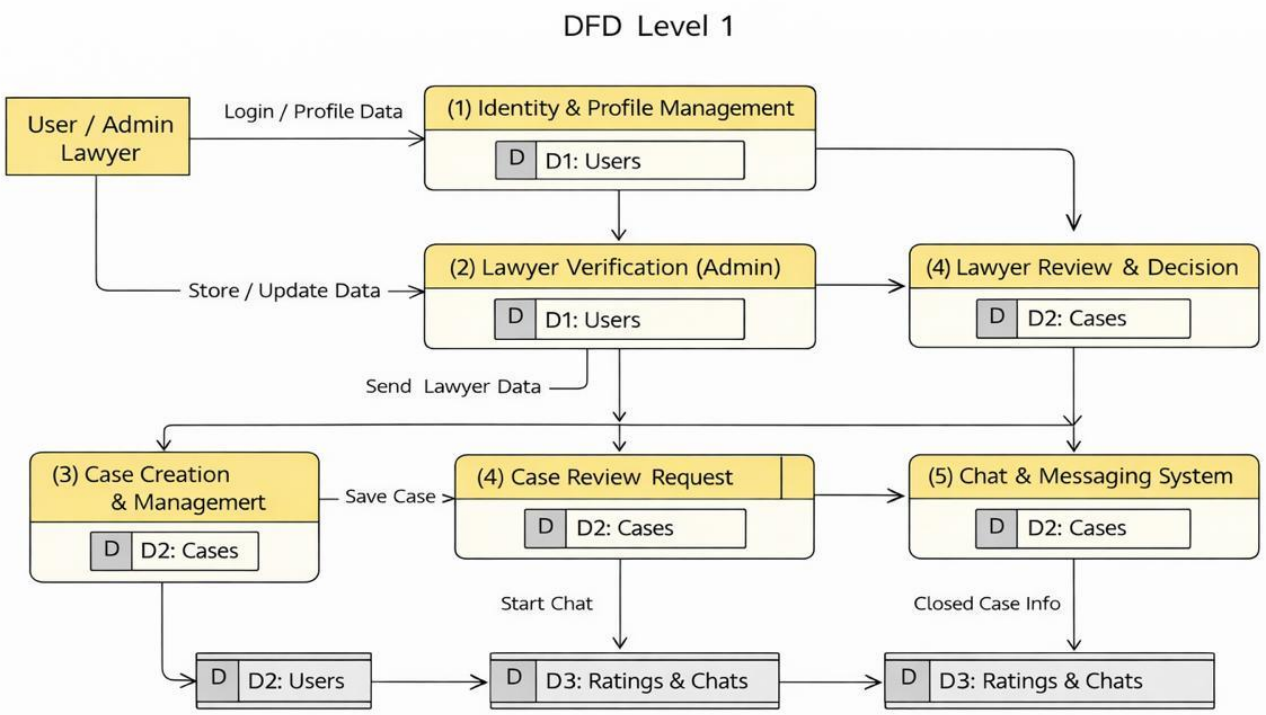


Figure 7. Data Flow Diagram (DFD) LEVEL 1

Chapter 4
Project Tools

Project Tools

Back-end:

C#:

- C# (pronounced “C-sharp”) is a modern, object-oriented programming language developed by Microsoft.
- It provides strong type-checking, garbage collection, and asynchronous programming features that improve performance and reduce runtime errors.
- C# is widely used for building web APIs, desktop apps, and enterprise-grade solutions, offering excellent integration with databases and cloud services..

.NET (ASP.NET Core):

- .NET is a powerful, open-source development framework created by Microsoft for building scalable, high-performance web and enterprise applications.
- It supports multiple programming languages, including C#, and offers built-in support for modern software architectures such as RESTful APIs, MVC, and microservices.
- ASP.NET Core, a cross-platform version of .NET, is optimized for cloud and container-based deployments, providing excellent speed, security, and modularity.

- Its dependency injection, middleware pipeline, and powerful API tools make it ideal for building secure and maintainable backend systems like LEX..

Entity Framework Core (EF Core):

- **Entity Framework Core** is an Object-Relational Mapper (ORM) that simplifies data access by allowing developers to interact with the database using C# objects instead of writing raw SQL.

- It supports **LINQ queries,** change tracking, migrations, and relationship management between entities.

- **EF Core** improves productivity and ensures consistency across data operations, making it an essential tool for LEX's database layer.

Identity and Authentication (ASP.NET Identity / JWT):

- ASP.NET **Identity** provides a complete authentication and authorization system for managing users, roles, and claims securely.
- When combined with **JSON Web Tokens (JWT)**, it supports token-based authentication for API access, ensuring that all communication within the LEX platform is secure and stateless.
- This setup enables features like email confirmation, password reset, role-based access, and multi-user session management.

AutoMapper:

- AutoMapper is a lightweight library that simplifies object-to-object mapping, commonly used to transform data between entities, DTOs (Data Transfer Objects), and view models.
- It reduces repetitive code and ensures clean separation between business logic and presentation layers.

MailKit / SMTP:

- **MailKit** is a modern, cross-platform .NET library used for sending and receiving emails.
- It supports SMTP, POP3, and IMAP protocols and is used within LEX for sending verification emails, password resets, and system notifications

Twilio:

- Twilio is a cloud communications platform that provides APIs for sending and receiving text messages, making phone calls, and other communication-related tasks. It's widely used for integrating SMS, voice, and other communication features into applications.

xUnit / NUnit:

- **xUnit** (or **NUnit**) are testing frameworks for .NET that support unit testing and integration testing.
- They ensure the backend logic works correctly, covering authentication, data access, and API response validation.
- Both frameworks integrate easily with CI/CD pipelines for automated testing and continuous quality assurance.

PostgreSQL:

- PostgreSQL is a powerful, open-source relational database management system (RDBMS). It's known for its reliability,

scalability, and support for advanced features like JSON support, full-text search, and extensibility through custom functions and extensions.

pgAdmin:

- pgAdmin is a popular open-source administration and development platform for PostgreSQL. It provides a graphical interface for managing databases, writing SQL queries, and performing tasks like backup, restore, and user management.

GitHub:

- A platform for version control and code collaboration using Git. It allows developers to share code, track changes, and manage project versions efficiently.

Visual Studio Code (VS Code):

- Visual Studio Code is a free, open-source code editor developed by Microsoft. It's highly customizable and supports a wide range of programming languages through extensions. Features like IntelliSense (code completion), debugging, Git integration, and a built-in terminal make it a popular choice for developers.

Front-end:

Mobile Application

Dart:

- Dart is a programming language developed by Google, primarily used for building mobile, web, and desktop applications. It's the language behind the Flutter framework and is known for its fast performance, strong typing, and just-in-time (JIT) and ahead-of-time (AOT) compilation. Dart is designed to be easy to learn and highly productive for developers.

Flutter:

- Flutter is an open-source UI software development kit (SDK) created by Google. It allows developers to build natively compiled applications for mobile, web, and desktop from a single codebase. Flutter uses the Dart programming language and provides a rich set of pre-designed widgets, making it easy to create visually appealing and high-performance applications.

Google Maps API:

- The Google Maps API is a collection of APIs provided by Google that allows developers to integrate Google Maps into their applications. It includes features like displaying maps, adding markers and providing location-based services. It's widely used in web and mobile applications for mapping, navigation, and geolocation features.

Postman:

- Postman is a popular tool for testing and interacting with APIs. It provides a user-friendly interface for sending HTTP requests, inspecting responses, and automating API testing. Postman also supports features like collections, environments, and collaboration, making it a go-to tool for developers and testers working with APIs.

Android Studio:

- Android Studio is the official integrated development environment (IDE) for Android app development. It's built on IntelliJ IDEA and provides tools for coding, debugging, testing, and profiling Android applications. Android Studio includes an emulator, a visual layout editor, and support for Kotlin and Java, making it the preferred choice for Android developers.

Visual Studio Code (VS Code):

- Visual Studio Code is a free, open-source code editor developed by Microsoft. It's highly customizable and supports a wide range of programming languages through extensions. Features like IntelliSense (code completion), debugging, Git integration, and a built-in terminal make it a popular choice for developers across various platforms and languages

WEB - React

React:

- React is an open-source JavaScript library developed by Facebook for building dynamic and responsive user interfaces. It's component-based, allowing developers to create reusable UI elements that efficiently update when data changes. React's virtual DOM enhances performance, and its unidirectional data flow ensures predictable behavior across complex applications.

Redux Toolkit:

- Redux Toolkit is the official, modern approach to managing state in React applications. It simplifies the setup and structure of Redux by providing built-in tools for creating slices, reducers, and asynchronous logic. It improves scalability and makes state management more efficient and less error-prone .

JavaScript:

Dynamic high-level programming language that is crucial to web development. Alongside HTML and CSS, it is essential to web applications and makes interactive web pages possible. Because it supports imperative functional and event-driven programming styles JavaScript is f lexible enough for front-

end and back-end development. Building testing and documenting APIs is made easier with Postman a collaboration platform for API development. It offers an intuitive user interface for submitting requests, reviewing answers and running tests automatically.

React Router DOM:

- React Router DOM is a routing library for React that enables navigation between different pages or views within a single-page application (SPA). It allows developers to create dynamic routes, nested layouts, and protected routes for user authentication, ensuring smooth and seamless navigation.

Axios:

- Axios is a promise-based HTTP client used to communicate with APIs. It simplifies sending requests and handling responses, supports interceptors for authentication, and can be easily integrated with Redux or React Query to manage API calls and global data states.

Formik & Yup:

- Formik is a React library that simplifies form creation and management by handling state, validation, and submission. Combined with Yup, a schema-based validation library, developers can easily build dynamic and secure forms with real-time error handling and validation rules.

Material UI / Tailwind CSS:

- Material UI and Tailwind CSS are popular styling solutions for React.
- **Material UI** offers a collection of pre-styled, accessible components that follow Google's Material Design guidelines.
- **Tailwind CSS** is a utility-first CSS framework that allows developers to design modern, responsive interfaces quickly with complete customization flexibility

Postman:

- Postman is a popular tool for testing and interacting with APIs. It provides a user-friendly interface for sending HTTP requests, inspecting responses, and automating API testing. Postman also supports features like collections, environments, and collaboration, making it a go-to tool for developers and testers working with APIs.

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UI/UX:

Figma:

- Figma is a cloud-based design and prototyping tool used for creating user interfaces (UI), user experiences (UX), and collaborative design projects. It allows designers to create vector graphics, wireframes, and interactive prototypes in real-time. Figma's standout feature is its collaborative nature, enabling multiple users to work on the same project simultaneously. It's widely used by design teams for its ease of use, cross-platform accessibility, and integration with other tools.

Other Tools:

ClickUp:

- ClickUp is a cloud-based productivity and project management tool designed to help teams and individuals organize tasks, projects, and workflows. It offers features like task management, time tracking, goal setting, document collaboration, and integrations with other tools. ClickUp is highly customizable, allowing users to create workflows tailored to their specific needs. It's widely used by teams of all sizes to improve productivity and streamline project management.

Chapter 5
System Design

System Design

- **Intro Screen**

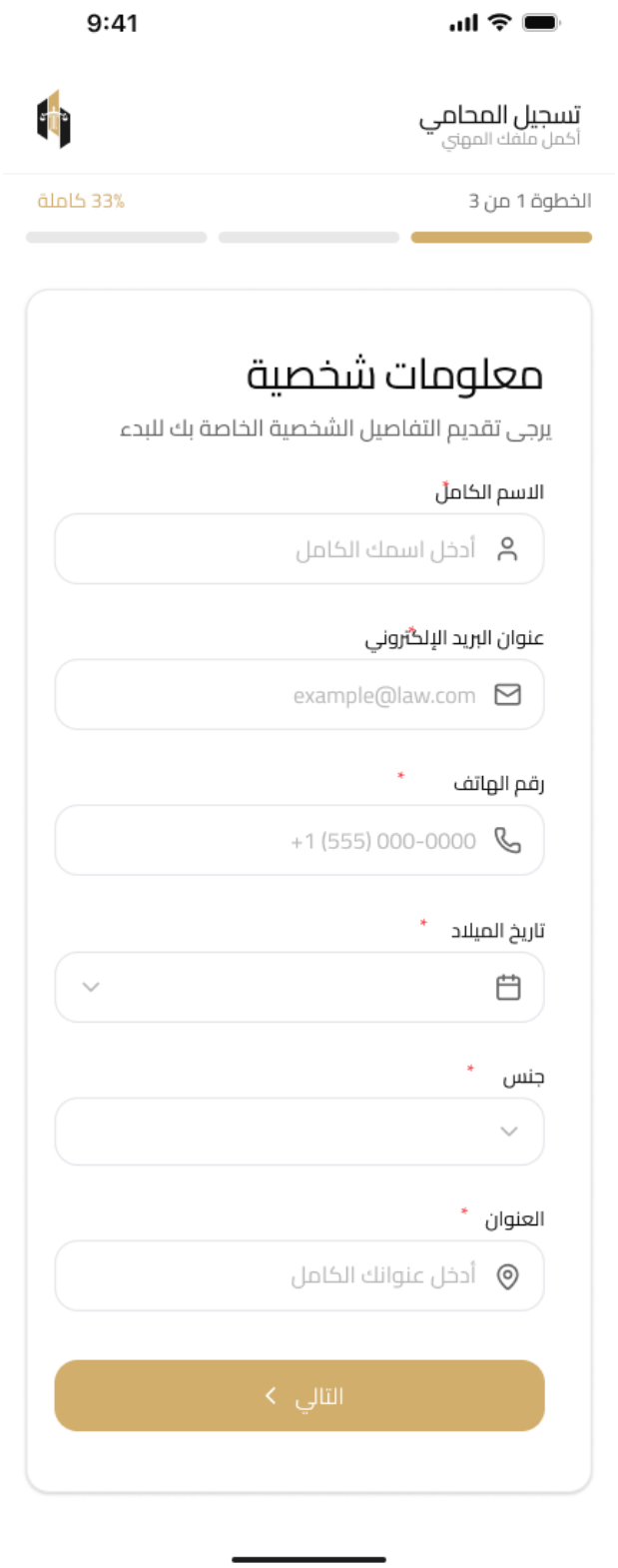




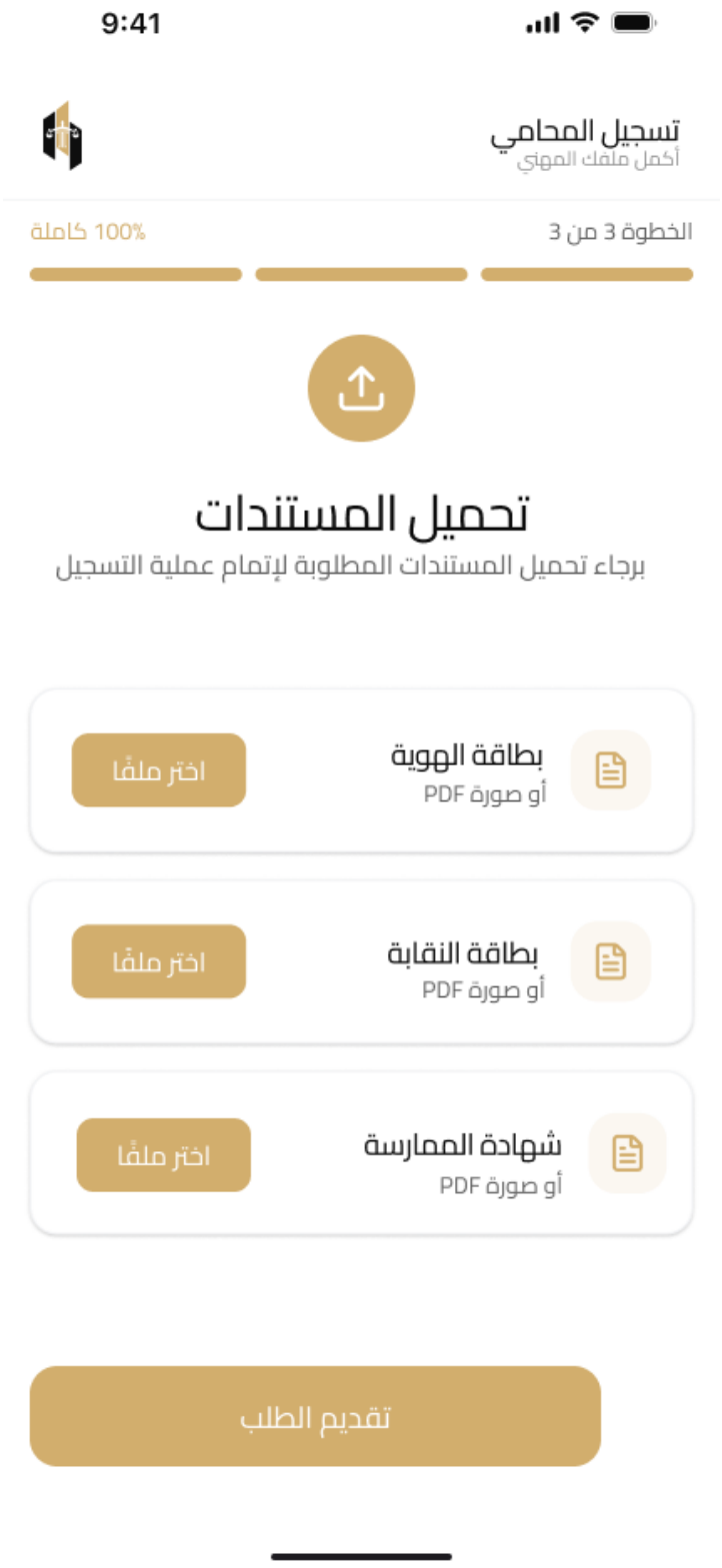
- **User Sign up**


9:41
أهلاً وسهلاً
الحصول على إدارة الخدمات القانونية الخاصة بك بسهولة
الاسم الكامل
Mohammed Ali
البريد الإلكتروني أو رقم الهاتف
shdbshdh2gmail.com
كلمة المرور
تذكرني
إنشاء حساب

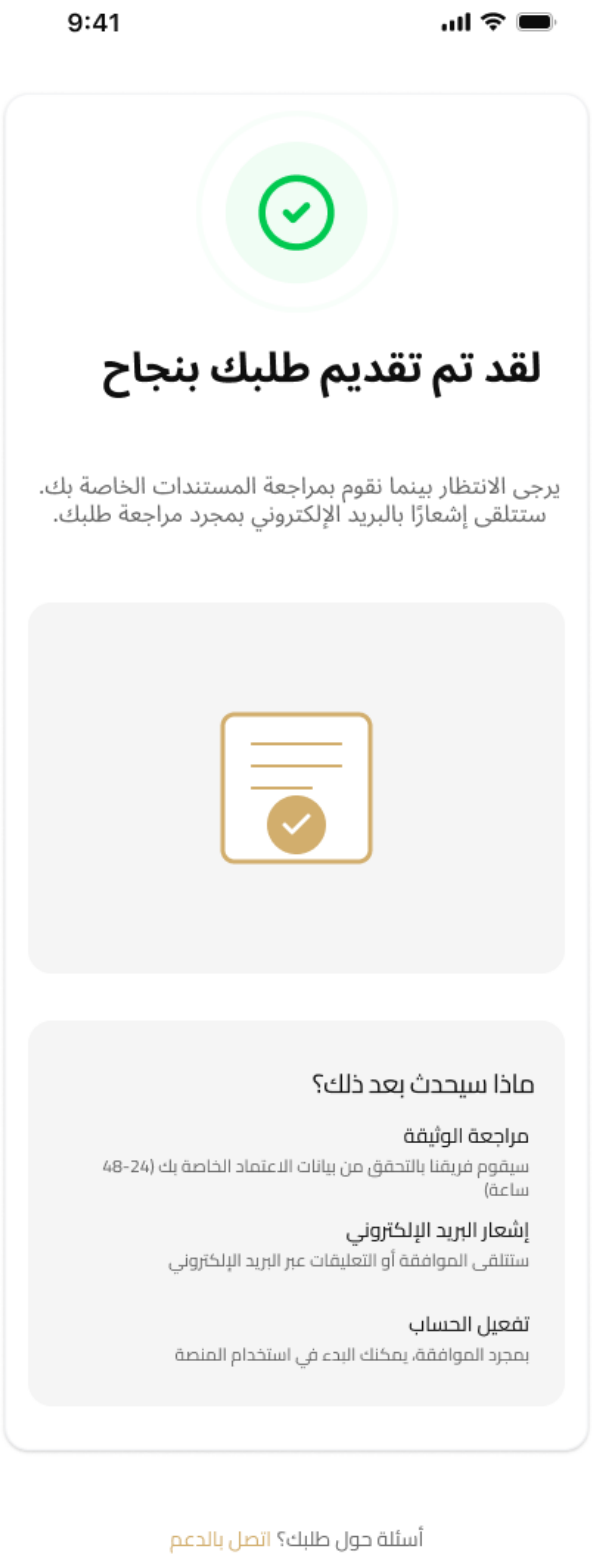
- **Lawyer Sign up**



- **Lawyer Sign up**



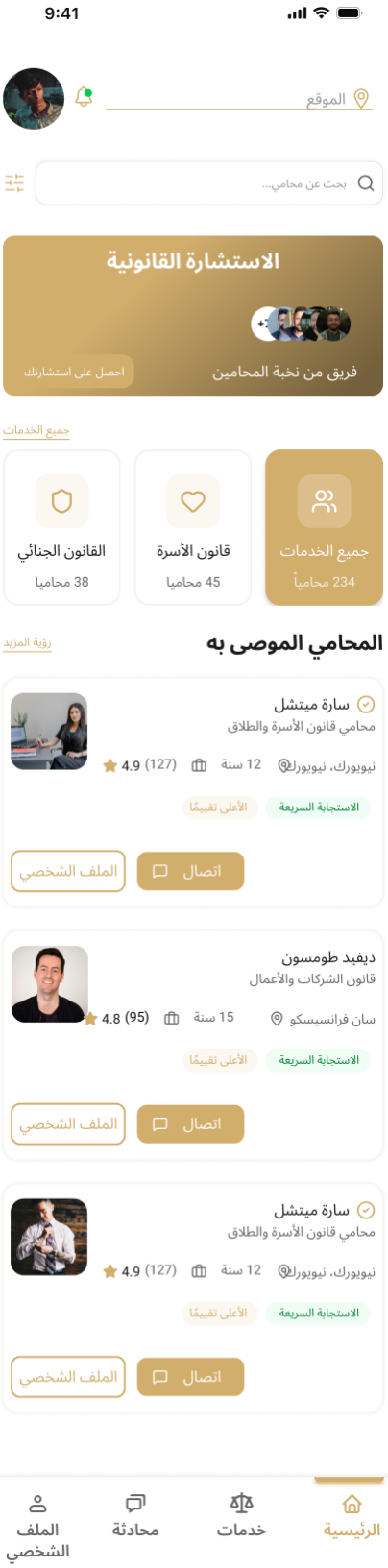
- **Lawyer Sign up**



- **Sign in**


9:41
قم بتسجيل الدخول لإدارة خدماتك القانونية بسهولة
مرحبًا بعودتك!
البريد الإلكتروني أو رقم الهاتف
shdbshdh2gmail.com
كلمة المرور
****
هل نسيت كلمة المرور؟
تذكرني
تسجيل الدخول
ليس لديك حساب؟ إنشاء حساب

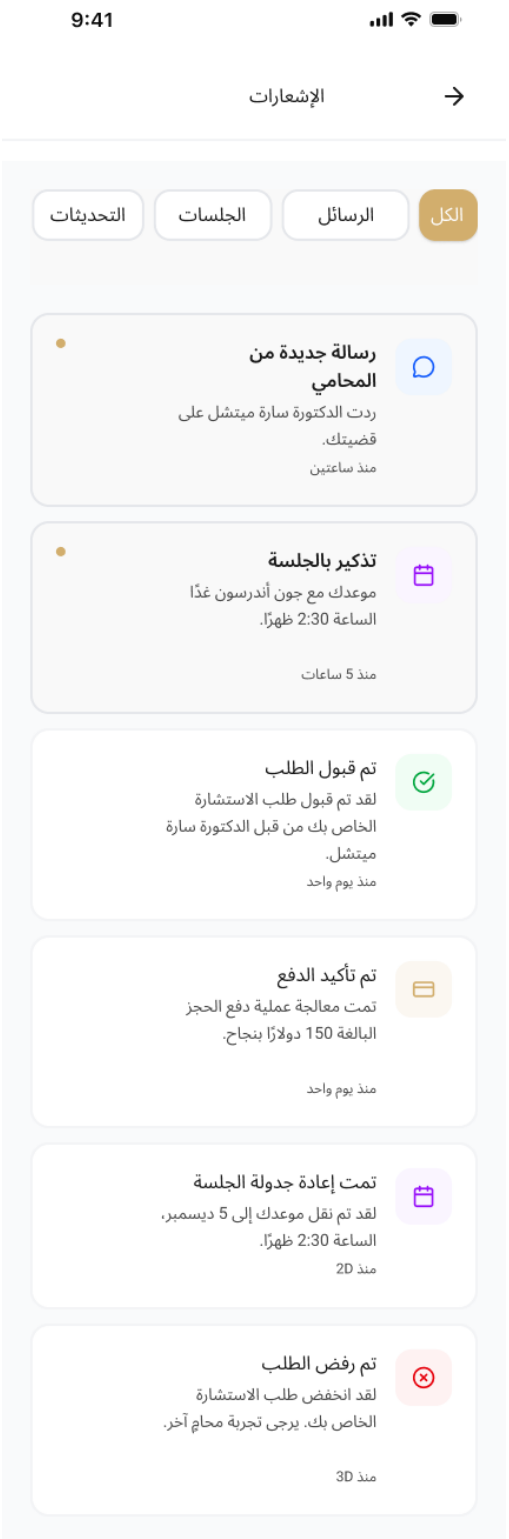
- **Main page**


9:41
الموقع
بحث عن محامي...
الاستشارة القانونية
فريق من نخبة المحامين
احصل على استشارتك
جميع الخدمات
جميع الخدمات
234 محامياً
قانون الأسرة
45 محاميا
القانون الجنائي
38 محاميا
المحامي الموصى به
رؤية المزيد
سارة ميتشل
محامي قانون الأسرة والطلاق
نيويورك، نيويورك
12 سنة
4.9 (127)
الاستجابة السريعة
الأعلى تقييمًا
اتصال
الملف الشخصي
ديفيد طومسون
قانون الشركات والأعمال
سان فرانسيسكو
15 سنة
4.8 (95)
الاستجابة السريعة
الأعلى تقييمًا
اتصال
الملف الشخصي
سارة ميتشل
محامي قانون الأسرة والطلاق
نيويورك، نيويورك
12 سنة
4.9 (127)
الاستجابة السريعة
الأعلى تقييمًا
اتصال
الملف الشخصي
الرئيسية
خدمات
محادثة
الملف الشخصي

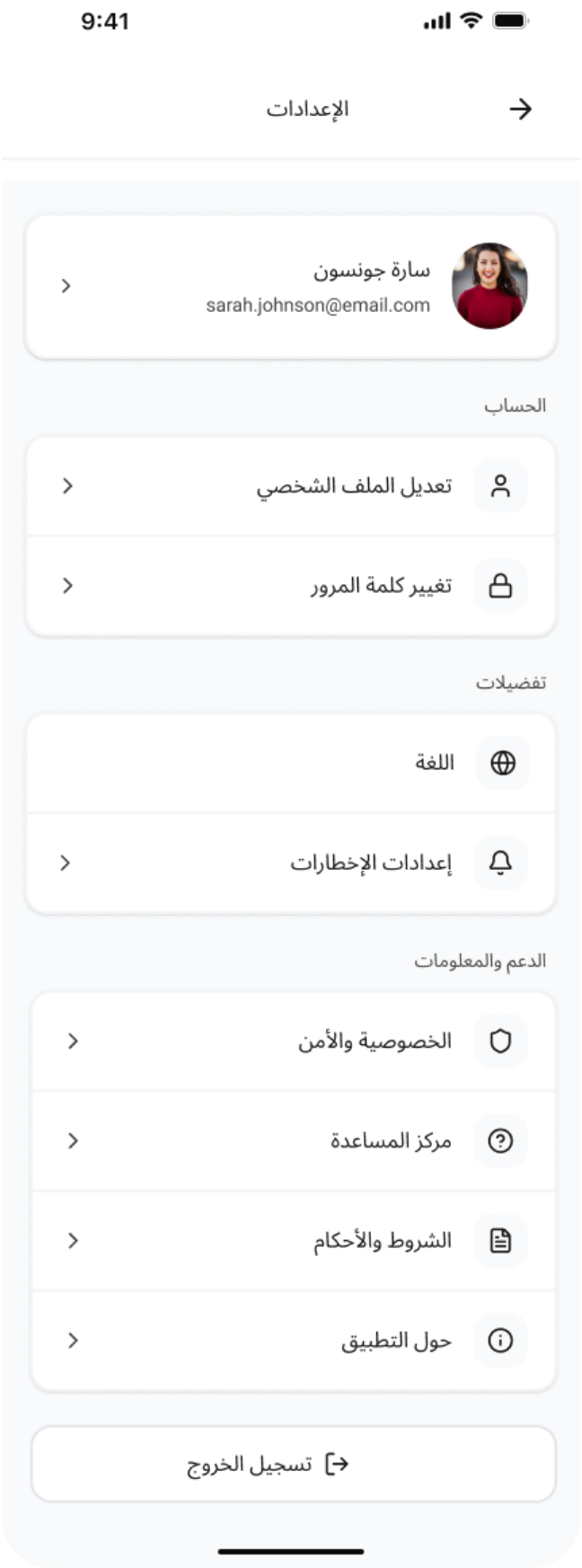
- **Chat**


9:41
سارة ميتشل
متصل
اليوم
مرحبًا! أحتاج إلى استشارة قانونية بخصوص نزاع عقدي. هل أنت متاح؟
10:30 صباحا
مرحبًا! نعم، أنا متاح. سأكون سعيدًا بمساعدتك في النزاع بشأن العقد. هل يمكنك تزويدي بمزيد من التفاصيل حول الوضع؟
10:32 صباحا
العقد_الاتفاق_2024.pdf
2.4 ميجابايت
أرفق وثيقة العقد لمراجعتك. تتعلق المشكلة بالبند 5.2 المتعلق بشروط الدفع.
10:35 صباحا
اكتب رسالة...

- **Notifications**



- **Settings**


9:41
الإعدادات
سارة جونسون
sarah.johnson@email.com
الحساب
تعديل الملف الشخصي
تغيير كلمة المرور
تفضيلات
اللغة
إعدادات الإخطارات
الدعم والمعلومات
الخصوصية والأمن
مركز المساعدة
الشروط والأحكام
حول التطبيق
تسجيل الخروج

Chapter 6
System Testing

1. User Registration (Sign Up)

- Test creating a new user account using email and password
- Verify email confirmation is required before login
- Check that user data (ID, email, password, full name) is saved in the database

Expected Result:
User account is created successfully and stored in database after email verification.

---

2. User Login

- Test login with correct email and password
- Test login with incorrect credentials
- Test login before email confirmation

Expected Result:
User can log in only after confirming email with valid credentials.

---

3. Lawyer Registration & Verification

- Test lawyer account creation
- Upload verification documents
- Admin reviews and approves lawyer account

Expected Result:
Lawyer cannot log in until admin approves documents

4. Role Access Control

- Test user access to user features
- Test lawyer access to lawyer features
- Test admin access to admin dashboard

Expected Result:

Each role accesses only its authorized functions.

---

5. Case Creation

- Test user creating a new legal case
- Enter case details correctly
- Save case data in database

Expected Result:

Case is created and stored successfully.

---

6. Sending Case to Lawyer

- Test sending case request to selected lawyer

Expected Result:

Lawyer receives the case request.

---

7. Lawyer Decision

- Test lawyer accepting a case
- Test lawyer rejecting a case

Expected Result:

Case status updates correctly in database.

8. Chat System

- Test opening chat after case acceptance
- Test sending and receiving messages
- Verify messages are stored in database

Expected Result:
Chat works correctly between user and lawyer.

---

9. Case Closing

- Test closing the case by user or lawyer

Expected Result:
Case status changes to closed.

---

10. Lawyer Rating

- Test rating lawyer after case is closed
- Attempt rating before closing case

Expected Result:
Rating allowed only after case is closed and saved in database.

---

11. Notifications

- Test sending notification when new case arrives
- Test notification when message received

Expected Result:
User receives real-time notification.

Chapter 7
Conclusion

Conclusion

In conclusion, the **LEX platform** represents a pivotal step toward the digital transformation of Egypt's legal ecosystem. By addressing long-standing challenges such as inefficient communication, limited transparency, and weak verification mechanisms, LEX provides a unified, secure, and user-centered environment that connects clients, lawyers, and law firms seamlessly. Through its integration of digital identity verification, encrypted communication, and real-time case management, the platform enhances trust, accountability, and operational efficiency within the legal sector.

Furthermore, by adopting modern technologies such as **.NET**, **React**, **Flutter**, and **PostgreSQL**, alongside agile development practices, LEX ensures scalability, adaptability, and long-term sustainability. Its multidisciplinary approach bridges the gap between technical innovation and social impact, fostering greater accessibility to legal services across both urban and rural regions of Egypt. Ultimately, LEX aims not only to modernize the delivery of legal services but also to promote a **transparent, efficient, and equitable justice system** that empowers citizens and legal professionals alike.

Chapter 8
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